

BCI Tasks Designs

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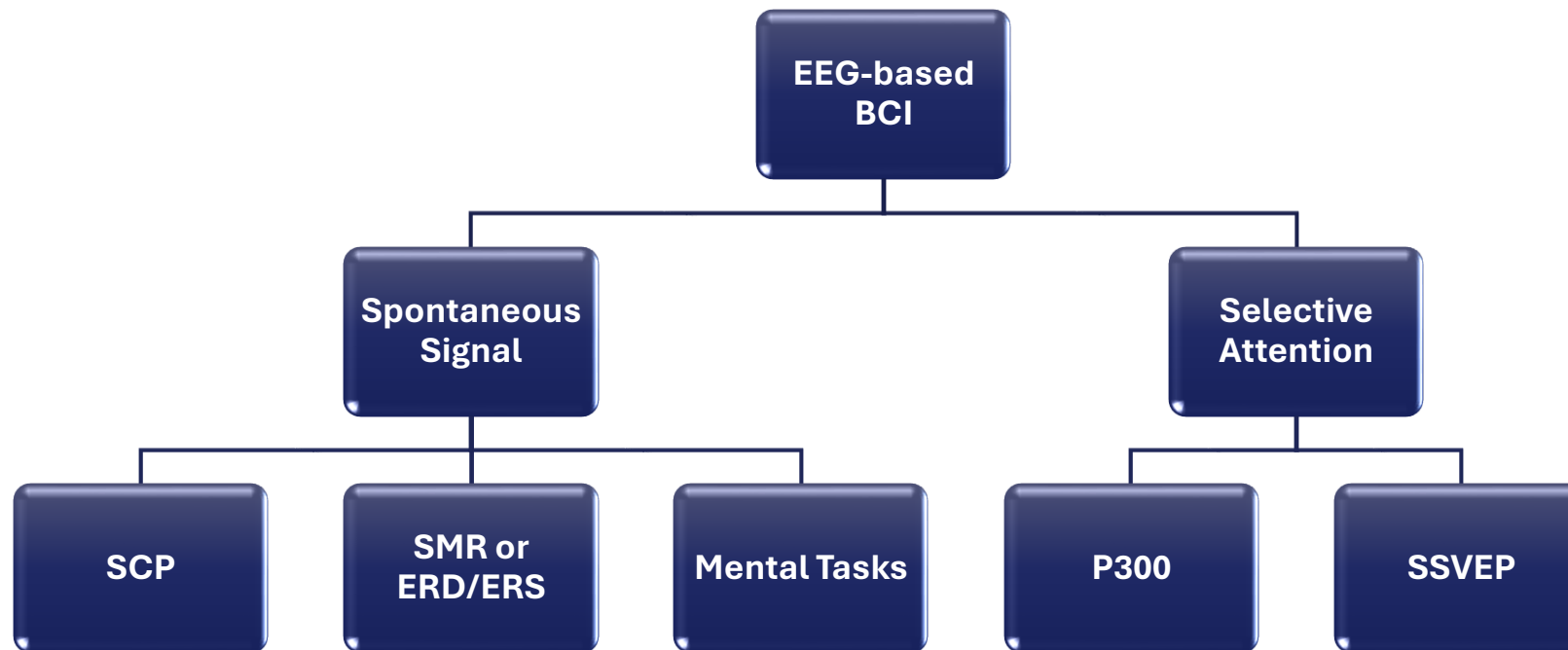
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Arab Academy for Science, Technology and Maritime Transport



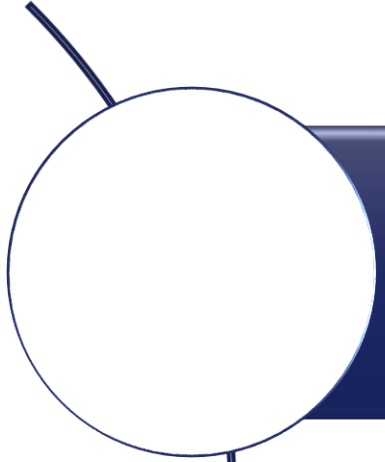
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EEG-based BCIs

EEG-based BCI systems can be designed to be based on selective attention or spontaneous mental signal.



EEG-based BCIs – Selective attention



The positive 300 ms (**P300**) and the steady-state visual evoked potential (**SSVEP**) are the examples of the selective attention method.



They are based on event-related potentials (**ERP**) and event-related oscillations (**ERO**) in the EEG, which are evoked by a specific stimuli.

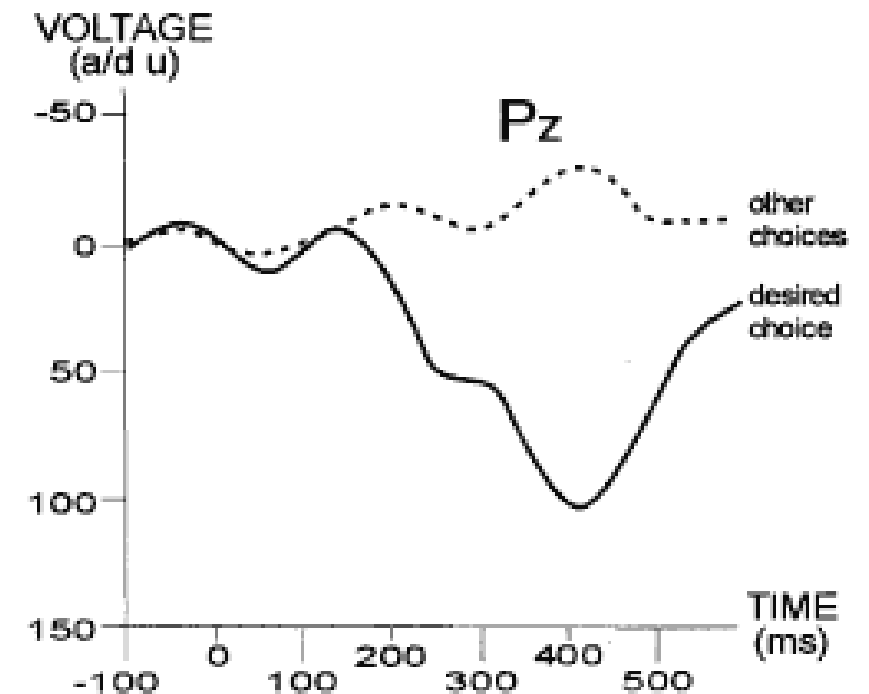
BCI signal – P300 component of the evoked potential

The P300 is a positive deflection in the recorded EEG after 300 ms following a stimulus and attenuate gradually after.

The paradigm enabling P300-based BCI to be used for communication and control has three attributes:

- Stimuli representing BCI outputs are provided in random order.
- Stimuli representing each conceivable output are delivered rarely.
- BCI users are requested to attend to the desired output stimulus.

P300 EVOKED POTENTIAL

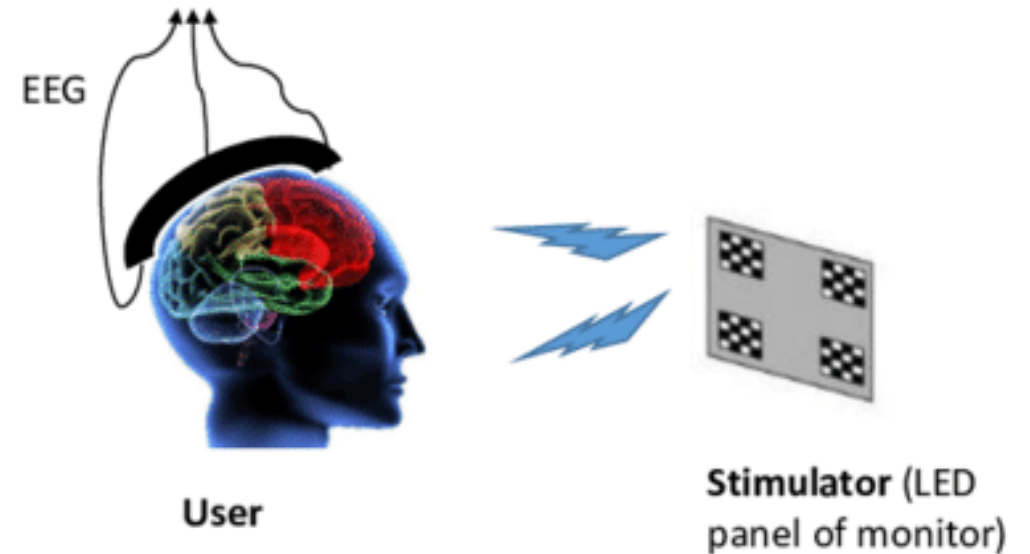


BCI signal – Steady-state visual evoked potential (SSVEP)

SSVEP uses external sudden visual stimuli such as a flash of light, an image, or a color or pattern shift.

The user is shown repetitive stimuli at a fixed frequency, where each stimulus is a BCI output.

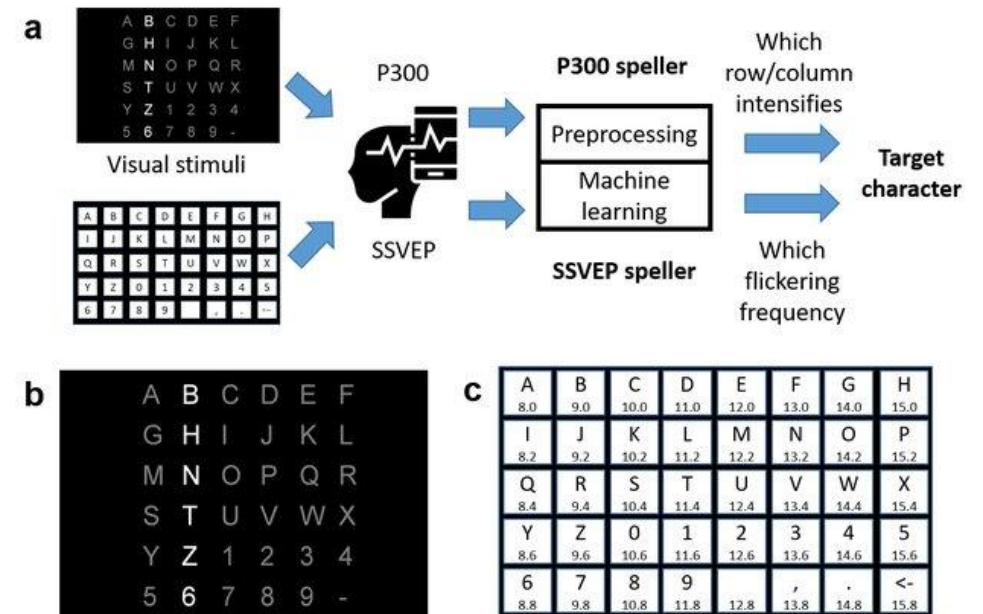
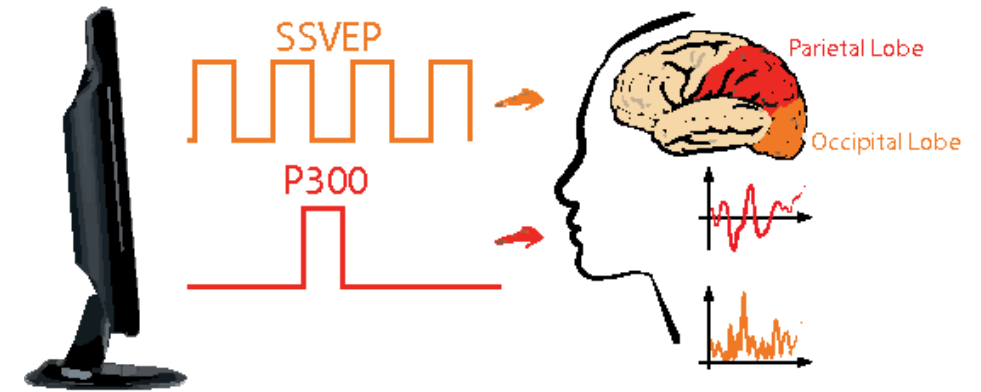
The user then selects the desired BCI output stimuli by locking eyes to the target for a window of time.



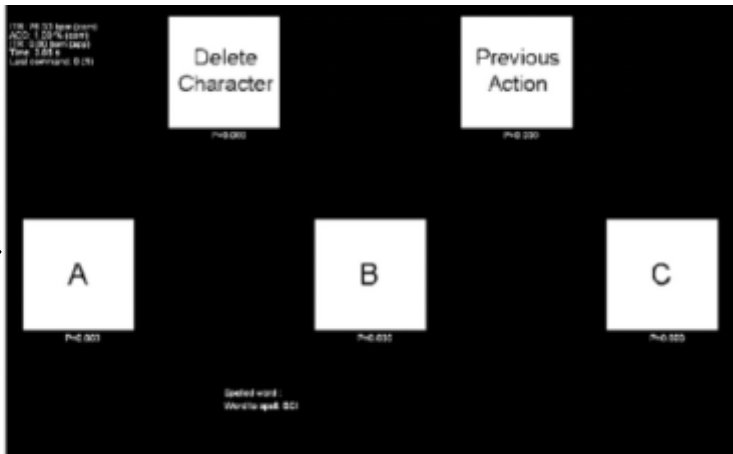
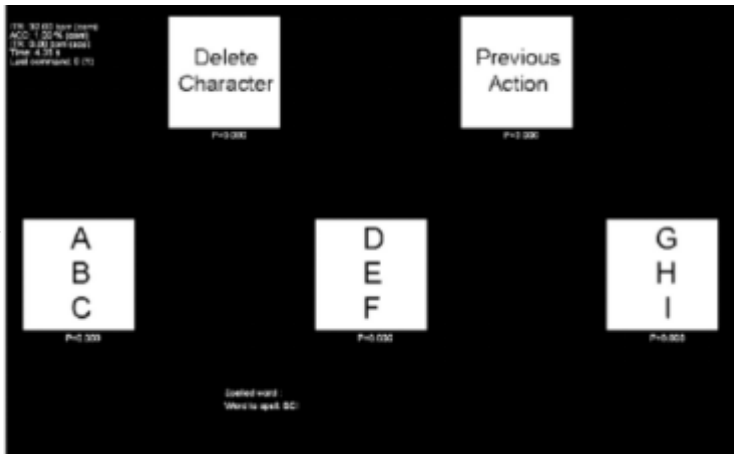
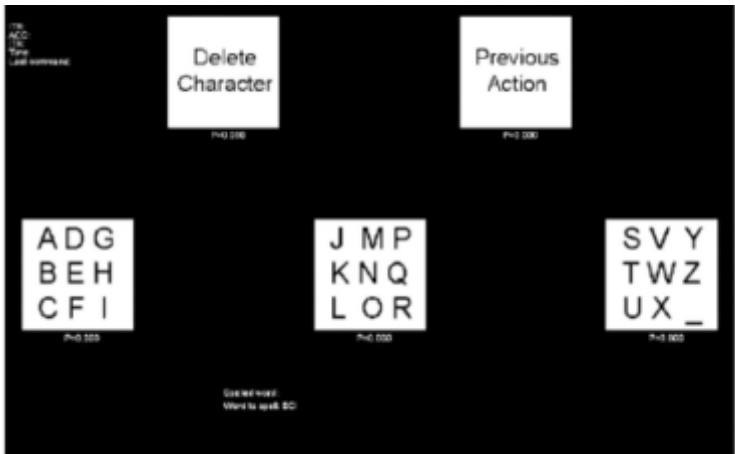
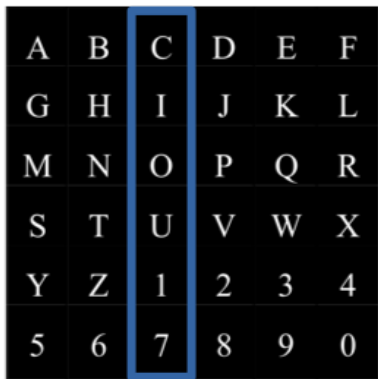
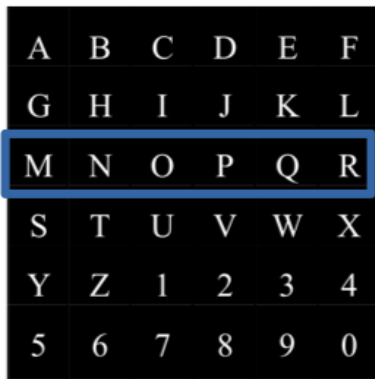
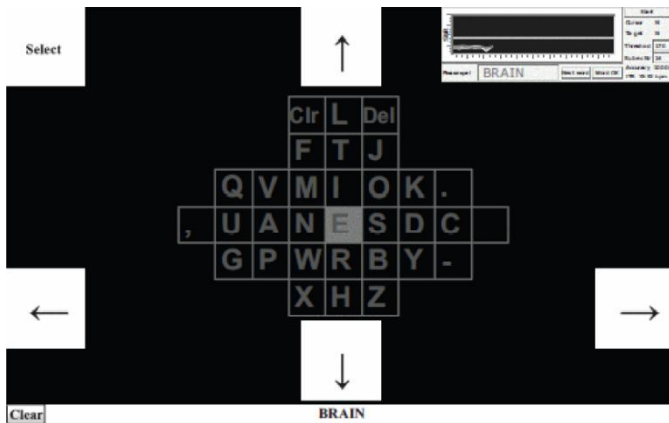
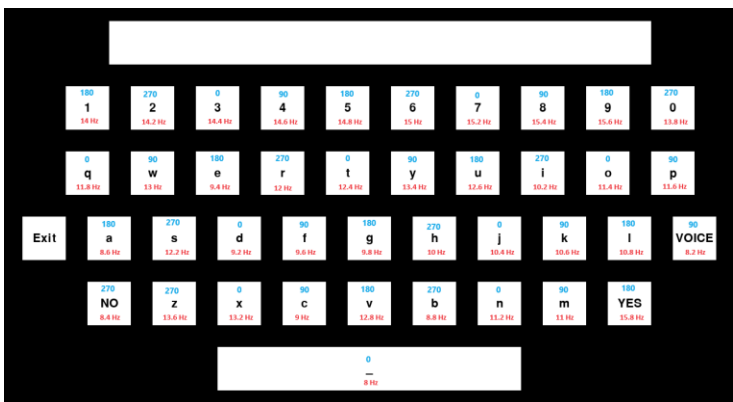
Selective attention experiment

Workflow of a P300 speller (top path) and an SSVEP speller (bottom path).

- For each speller, the subject watches the stimulation interface, focusing on the character he/she wants to input, and EEG signals are recorded and analyzed by the speller program.
- The P300 speller first identifies the row and the column that elicit the largest P300 and then outputs the character at their intersection.
- The SSVEP speller identifies the output character directly by matching the user's EEG oscillation frequency with the flickering frequency of each candidate character.



Experiment – Different speller paradigms



EEG-based BCIs – Spontaneous signal



BCI systems relying on spontaneous mental signals generated **voluntarily** by the user are based on **self-regulation**.

They include the slow cortical potential (**SCP**), sensorimotor rhythm (**SMR**) of motor imagery (**MI**) task based on event-related desynchronization/synchronization (**ERD/ERS**), and other mental tasks or **non-motor imagery mental tasks**.

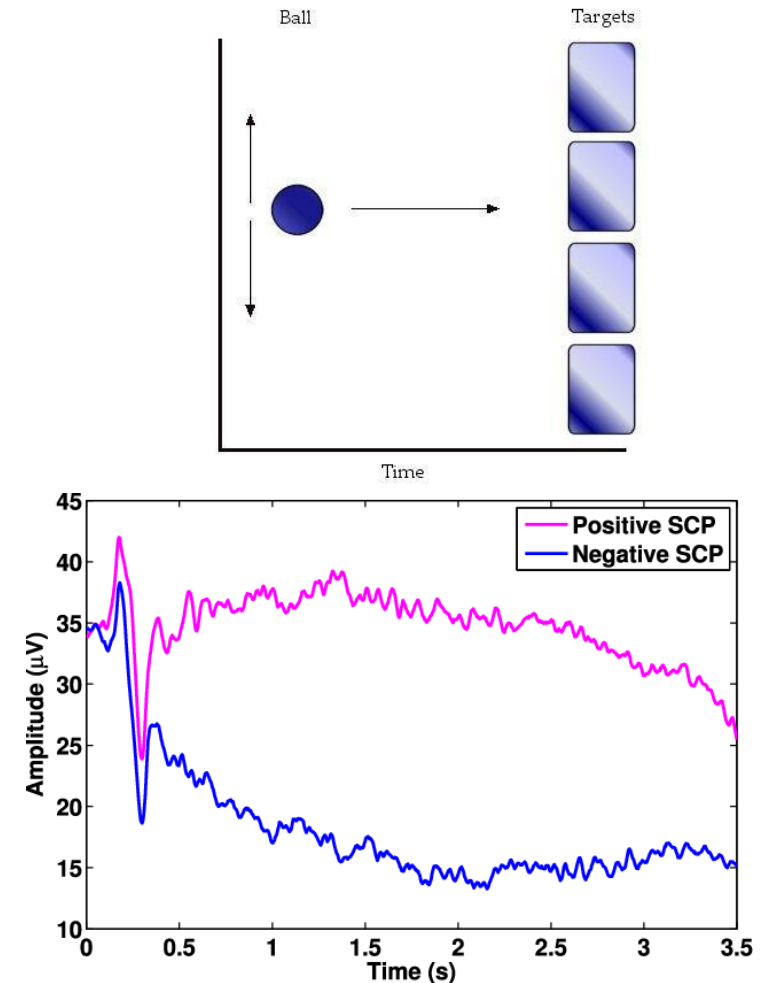
BCI signal – Slow cortical potential (SCP)

Slow, multi-second shifts in the brain's baseline voltage (Negative = "Ready/Active", Positive = "Relaxed/Idle").

Users can learn to shift this voltage on command, but it requires extensive neurofeedback training.

SCPs also happen naturally when the user anticipates a cue or prepares for a physical movement.

Requires clinical-grade equipment; consumer BCI headsets filter out these slow signals completely.



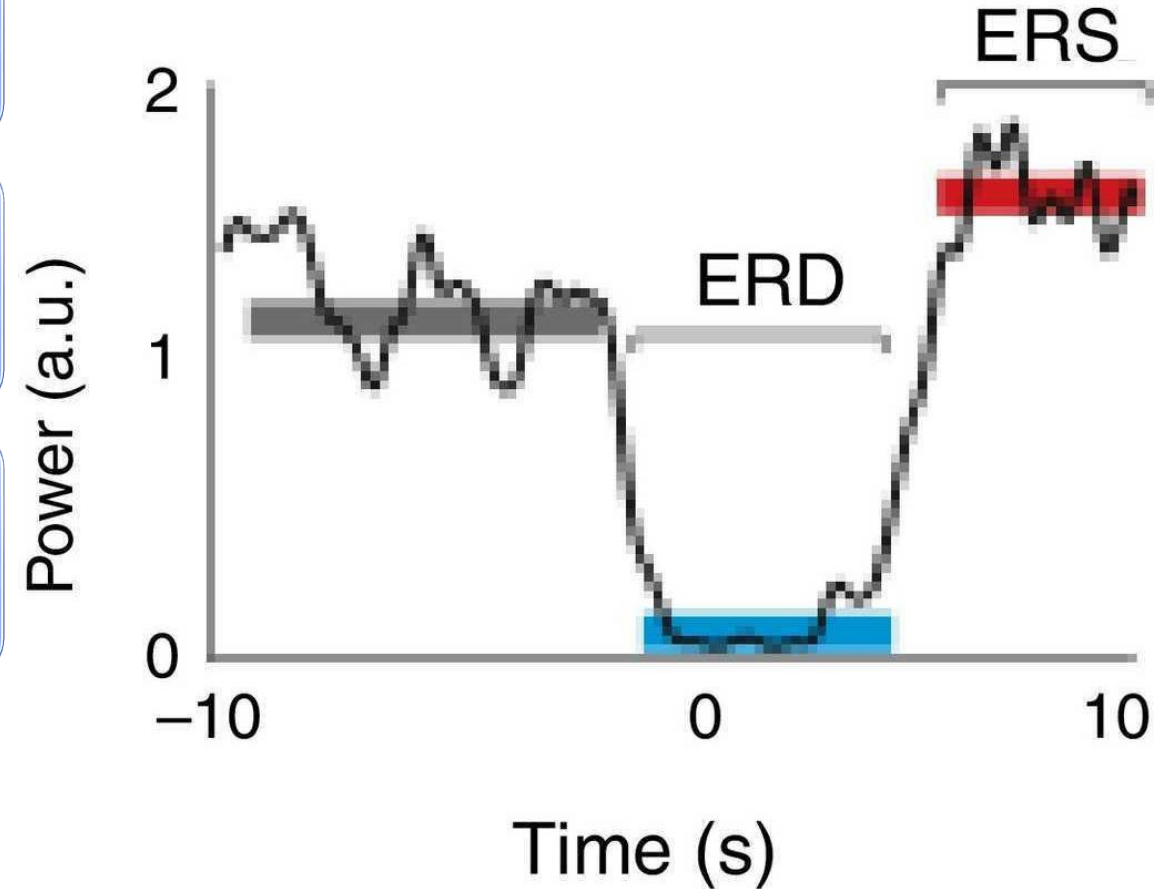
BCI signal – Sensorimotor rhythm (SMR)

BCIs based on SMR uses MI task associated with the oscillatory EEG signals from the sensorimotor cortex.

The SMR uses three major frequency bands: mu (8-14 Hz), beta (14-30 Hz), and gamma (30-200 Hz).

ERD and ERS patterns are basically associated with actual or imagined movements, such as foot, leg, hand, or tongue movements.

- At the time of the movement onset, it can be found that SMR decreases, known as event-related desynchronization (**ERD**).
- When the movement is complete, it can be found that SMR increases, known as event-related synchronization (**ERS**).

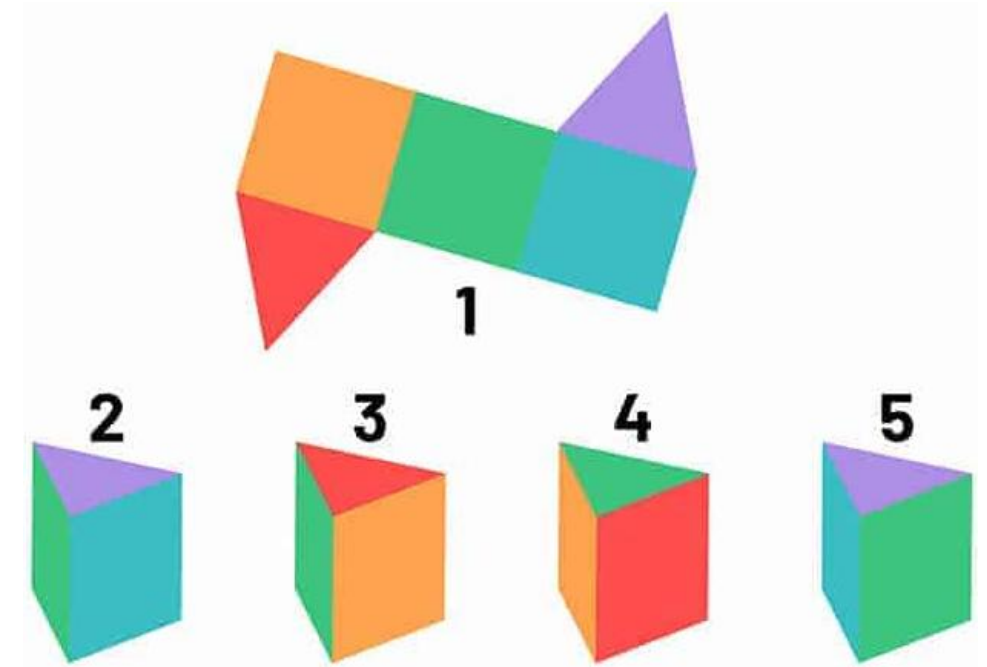


BCI signal – Non-motor imagery mental tasks

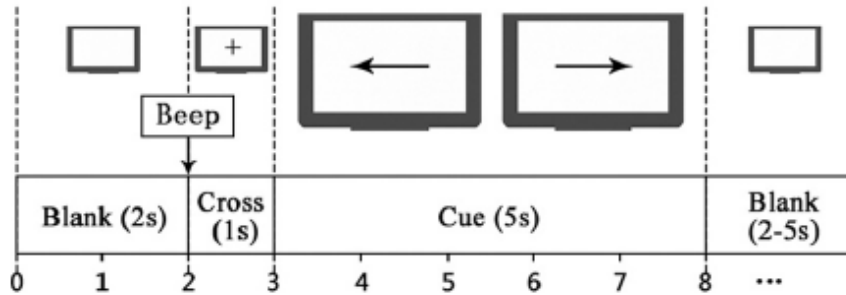
Although SMR is a common method employed by many research publications, there are still patients who cannot use it.

Using mental tasks rather than MI would be a better method. The tasks investigated include:

- **Relaxation (baseline)**, mental multiplication, mental geometric figure rotation, mental letter composition, and mental visual counting.



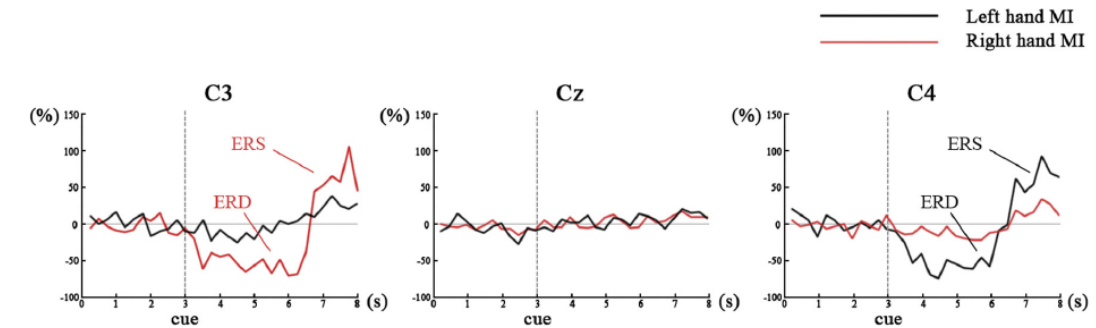
Spontaneous signal experiment



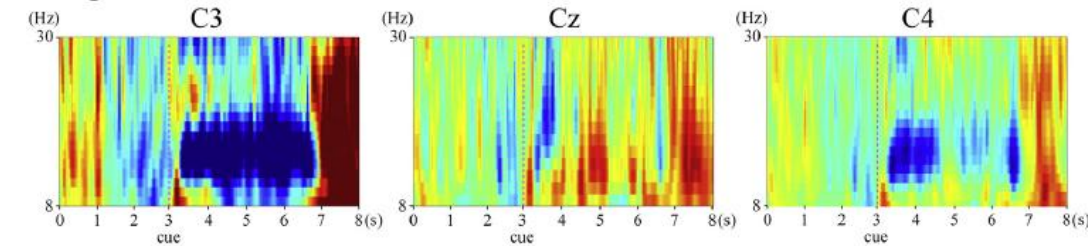
Subjects were instructed to perform two MI classes, i.e., left-hand or right-hand movements.

Each subject completed many trials, consisting of random sequences of trials of two classes of MI.

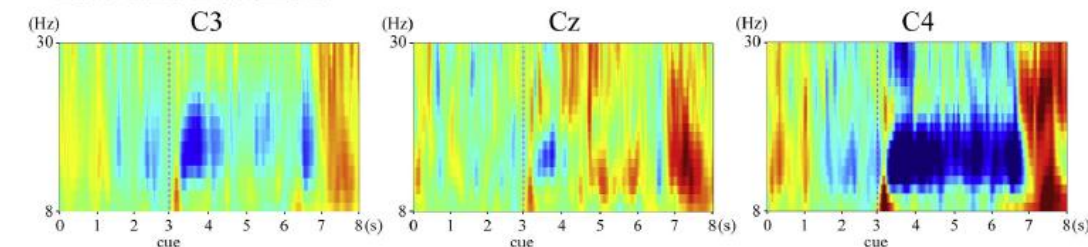
- Each trial started with the presentation of an acoustical warning tone and a cross.
- One second later, a cue was randomly chosen in one of two (“←” and “→”), which indicated the movement to be imagined.
- After the cue was presented, subjects had to perform the MI for 5 s, until the screen content was erased.
- After a short pause (random duration between 2 and 5 s) the next trial started.



•Right hand movement



•Left hand movement



Comparison EEG-based BCI methods

BCI-EEG Method	Advantages	Disadvantages	Recommended Applications
P300	- Feature as a positive shift in the EEG about 300 ms after the stimulus. - No requirement of significant training.	- Need external stimulus. - Highly affected by the background noise. - Patients with attention lapses were reported not being able to use it.	- Speller Programs - Home Automation - Computer games
SSVEP	- Feature in specific frequency in response to a visual stimulus modulated. - No requirement of significant training.	- Need external stimulus. - Highly affected by the background noise. - Patients with attention lapses were reported not being able to use it.	- Speller Programs - Home Automation - Computer games
SCP	- Feature as negative and positive shifts (Identifiable pattern). - Can enhance other BCIs as hybrid systems.	- Require a long time for training (the longest). - Cannot do multidimensional control	- Computer games - Hybrid BCIs
SMR	- Feature as ERS/ERD ratio (Identifiable pattern). - No need for external stimulus.	- Require a long time for training. - Patients with motor cortex dysfunction were reported not being able to use it.	- Computer games - Wheelchair control - Prostatic devices control
Mental task	- Non-motor imagery tasks. - No need for external stimulus. - Alternative option of disadvantages of SMR.	- Require a long time for training based on the task. - User needs to be familiar with the mental task.	- Home Automation - Computer games - Wheelchair control

Practical notes

For the instrumentation point of view, portability, convenience and cost effectiveness are important aspects for a practical BCI system.

A **lower number of EEG channels** used would also **improve portability** and convenience aspects for practical BCI.

We must ensure, however, that **this does not degrade overall system performance.**

Experimental approaches to study human brain

Controlled experiments

The study questions determine the **stimuli and experimental circumstances**, which must include baseline and task conditions.

All investigations must be able to statistically compare neural activity to a baseline condition.

Behavior-driven experiments

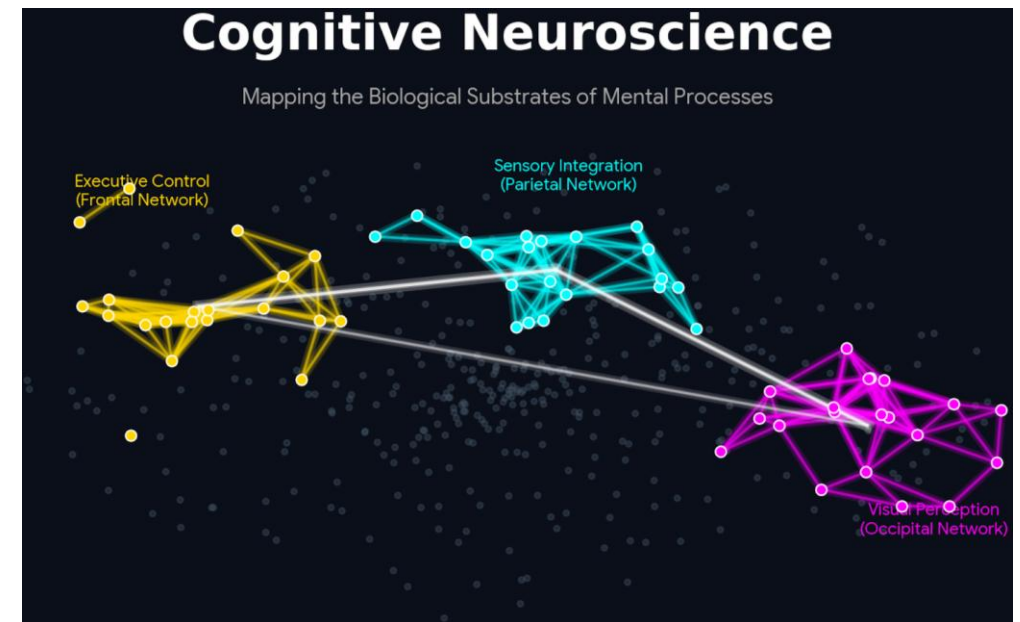
A well-designed task engages participants in a **specific behavior or cognitive process**.

The collected neuroimaging data is analyzed to identify regions of the brain that are activated during the task and to correlate brain activity with behavioral performance.

Cognitive neuroscience

Every abstract thought you have, like focusing intently on a puzzle or performing mental math, **generates a physical, measurable electrical field** across your cortex.

Cognitive neuroscience gives us the exact **framework to read these electrical fields and map them to specific mental states**, like relaxation or high cognitive workload.



What is attention?

Attention is a cognitive function that is activated at every single consciousness moment.

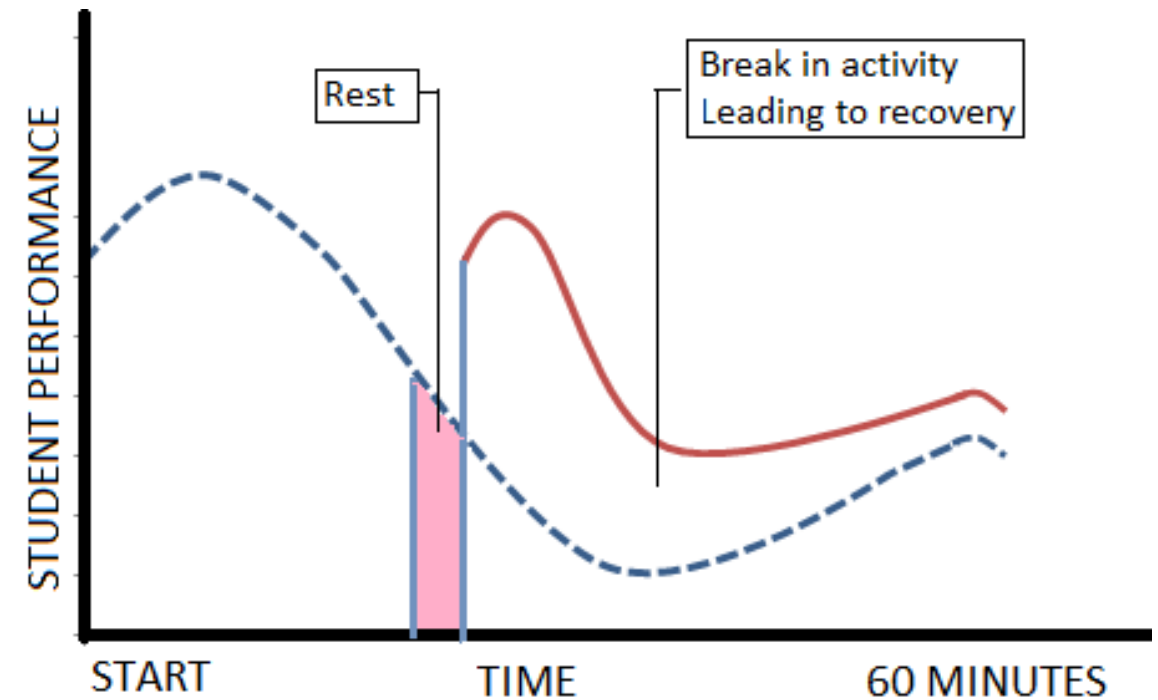
Attention simply allocates cognitive resources to set a priority based on the incoming information.

Attention decline

Subjects when observing a scene and ignore irrelevant stimuli for a long period, the ability to focus weakens.

The attention drop is a trend line because attention fluctuates over time.

When investigating performance fluctuations, consider arousal, attentional allocation, attentional effort, and attentional cost.



Theories to explain attention decline

Mindlessness theory

- Mindlessness (**boredom**) theory assumes vigilance decrease is due to the absence of externally supported attention as vigilance task is monotonous.
- To stay awake, the subject must sustain internally-driven attention.

Resource-depletion theory

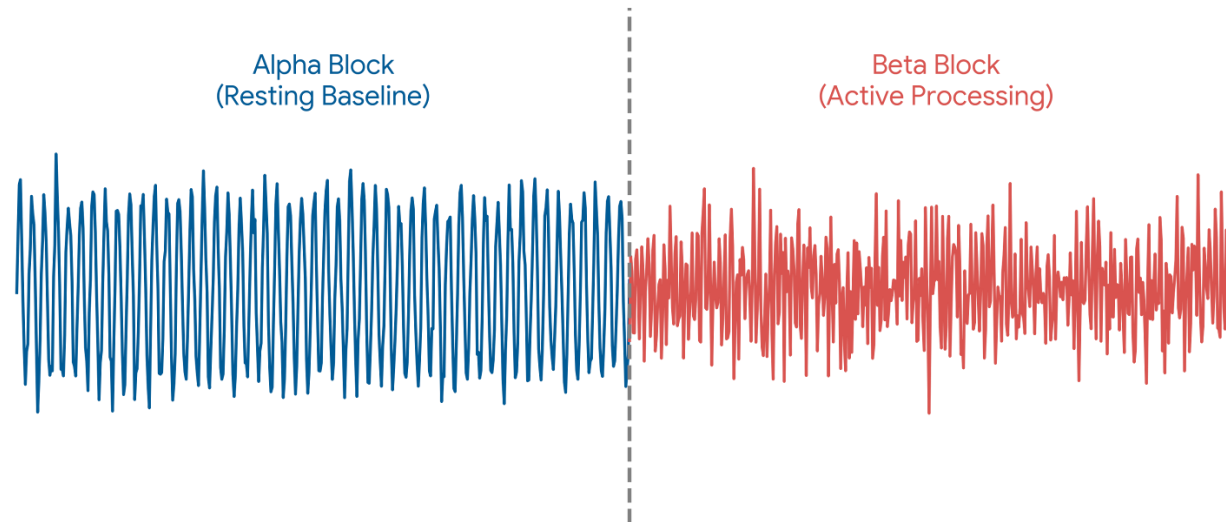
- Resource depletion (**mental fatigue**) theory suggests that lack of vigilance is due to inadequate cognitive resources for information processing.
- The subject must be always active to distinguish significant from irrelevant stimuli under uncertain conditions, leaving no time for rest.

The theory of cognitive states

To build a focus tracker, we are primarily concerned with two frequency bands. The core ideology behind our cognitive task is contrast.

Alpha is your baseline; it is the brain's default resting state, while **Beta is your active processing state.**

We need to design an experiment that captures a pure, uninterrupted block of Alpha, followed by a pure, high-intensity block of Beta.



Finding your brain's zero: The resting baseline

There is no universal "normal" for human brain activity. Your resting brainwaves are as unique to you as your fingerprint.

Before the computer can detect when your brain is working hard, it first needs to learn exactly what your brain looks like when it is doing absolutely nothing.

Closing your eyes naturally triggers a distinct, relaxed brainwave state. Alternating between eyes open and closed gives the system a clear, high-contrast anchor point to calibrate against.

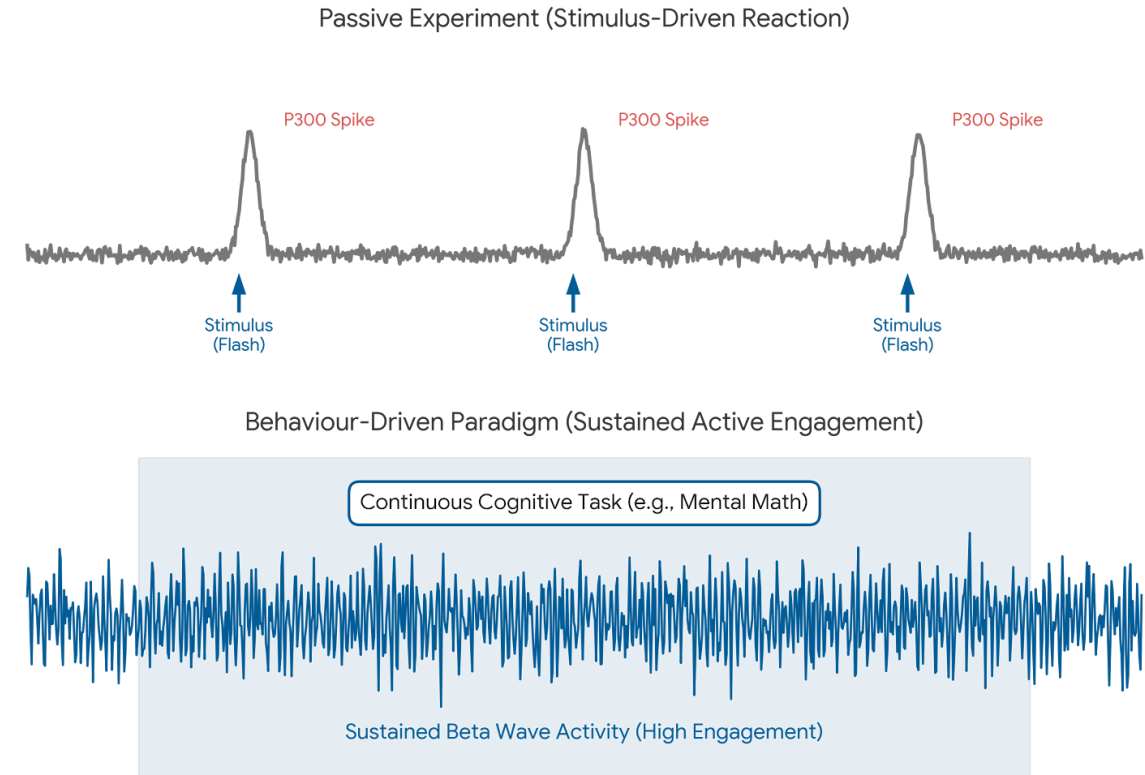
By recording this resting state, the system creates a custom baseline just for you. This is what allows it to accurately detect the exact moment your mind shifts from "rest" into "active focus" during the task.

Behaviour-driven task design

Unlike the P300 Speller, which is a highly controlled, passive experiment, the paradigm here is behavior driven.

We are not waiting for the brain to react to a stimulus; we are forcing the brain into a sustained state of work.

The ideology here draws on established neuroscientific research standards: to measure focus, you must design an environment where the subject has no choice but to remain highly engaged for the entire duration of the trial.



Hardware architecture: Electrode mapping

The sensor constraint

- The Muse 2 does not possess occipital or motor strip electrodes. SSVEP and MI paradigms are physically impossible with this headset.

Electrode configuration (Standard 10-20 system positioning)

- **Prefrontal channels:** AF7, AF8 (Capture frontal beta rhythms and cognitive focus).
- **Temporal channels:** TP9, TP10 (Capture lateral alpha rhythms and relaxation).
- **Right-ear reference:** CMS/DRL passive grounding system.

Physiological artifacts and noise control

The amplitude reality

- Cortical EEG signals are extremely weak (10 to 100 microvolts).
- Muscle signals (EMG) are massive (millivolt scale) and easily drown out brainwaves.

Primary noise vectors on the Muse 2

- **Eye blinks (EOG):** Generates massive low-frequency spikes on the frontal AF7/AF8 sensors.
- **Jaw clenching (EMG):** Floods the temporal TP9/TP10 sensors with high-frequency noise.

Strict experimental posture controls

- Feet flat on the floor, hands resting comfortably on the thighs.
- Head resting straight, avoiding micro-movements or looking around.
- Jaw completely slack (slightly open) to fully eliminate clench artifacts.

Muse 2 offline pipeline

Cognitive tasks design

Required libraries and dependencies

os, sys, time, csv, random	Used for directory management, system exits, protocol timing, writing the recorded EEG data to flat files, and generating randomized math problems.
tkinter	Builds the <u>initial pre-registration window</u> to capture participant details and setup secure saving folders before the main system launches.
threading, multiprocessing	Vital for splitting the workload across different CPU cores, ensuring the background Bluetooth stream, the operator dashboard, and the participant UI run simultaneously without blocking each other.
numpy, scipy.signal, collections.deque	Utilized for high-speed array operations, executing causal bandpass/notch filters, extracting power spectral density (PSD) features, and managing sliding time windows.
pygame, winsound	Drives the dual-screen graphical interfaces and triggers precise auditory beeps for protocol phase transitions.
pylsl, muselsl	Bridges the direct Bluetooth Low Energy (BLE) connection to the headset and handles the LSL network broadcasting.

Dual-screen BCI paradigm pipeline overview

Step 1: Pre-registration UI

- A graphical form to capture participant **metadata** and explicitly select the targeted cognitive stimulus (**Math vs. Words**) before launching the system.

Step 2: Multiprocessing initialization

- The core launcher that establishes shared memory variables and spawns isolated CPU processes.

Step 3: Background hardware streamer

- The daemon process securing the raw hardware link to the headset.

Step 4: Operator diagnostics view (Screen 1)

- A dedicated control panel for the **researcher** to monitor signal quality and cognitive states in real-time.

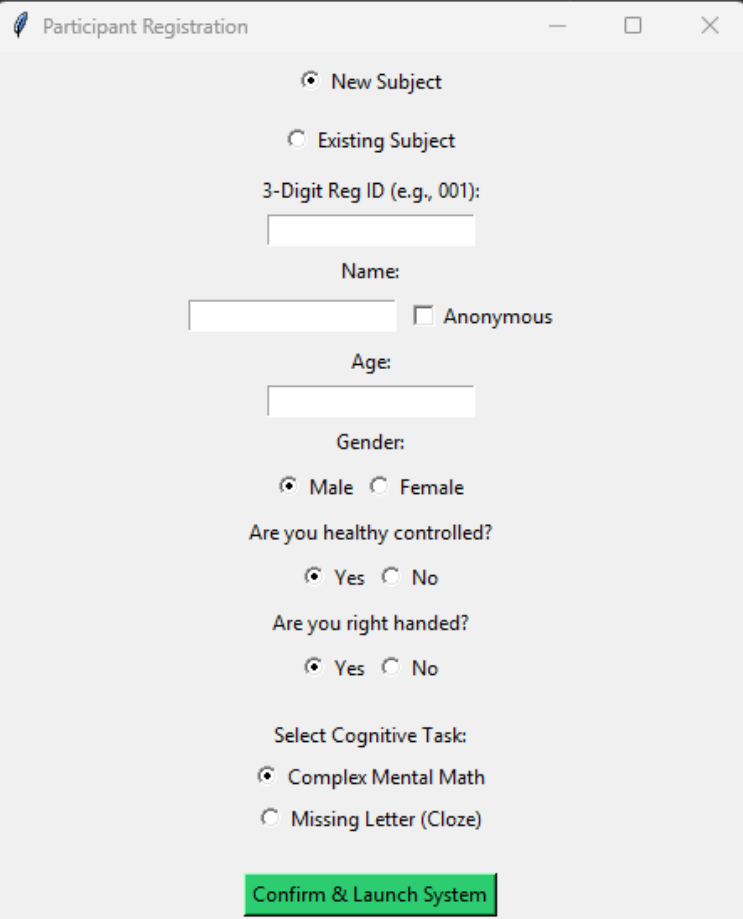
Step 5: Participant protocol (Screen 0)

- The primary task engine that dynamically generates the chosen cognitive task, walks the **subject** through baselining, and logs the unified data.

Pipeline mapping - Setup and architecture

Step 1: Pre-registration UI (*run_pre_registration*)

- **Functionality:** Generates a Tkinter window with radio buttons to toggle between "New" and "Existing" subjects, alongside a new toggle to select the designated Cognitive Task (Complex Mental Math or Missing Letter).
- **Code mapping:** Evaluates a 3-digit ID input to automatically generate the subject's profile, explicitly builds and targets the /offline/ subfolder (`os.makedirs(offline_folder_path)`), logs metadata into a `personal_info.txt` file, and passes the selected `task_type` variable to the downstream protocol engine.



The screenshot shows a Tkinter window titled "Participant Registration". It contains the following elements:

- Two radio buttons: "New Subject" (selected) and "Existing Subject".
- A text input field for "3-Digit Reg ID (e.g., 001):".
- A text input field for "Name:".
- An "Anonymous" checkbox.
- A text input field for "Age:".
- A "Gender:" section with "Male" (selected) and "Female" radio buttons.
- A question "Are you healthy controlled?" with "Yes" (selected) and "No" radio buttons.
- A question "Are you right handed?" with "Yes" (selected) and "No" radio buttons.
- A "Select Cognitive Task:" section with "Complex Mental Math" (selected) and "Missing Letter (Cloze)" radio buttons.
- A green "Confirm & Launch System" button at the bottom.

Why do we collect profile data?

The brain divides heavy workloads, like math and language, between the left and right hemispheres based on your **dominant hand**. This tells our algorithm exactly which side of your head to monitor.

Your resting brainwave frequency and cognitive processing speeds naturally change over time. **Age** data helps us calibrate the system to your personal biological metronome.

Natural variations in anatomy (like skull thickness) and baseline frequencies affect the raw strength of the EEG signal. This allows us to normalize the data accurately across the entire group of **genders**.

Brainwaves are highly personal biometric data, and you have the right to keep them private. More importantly, hiding your **name** eliminates "test anxiety"—keeping you relaxed prevents stress-induced muscle tension from ruining the delicate EEG signals!

Pipeline mapping - Setup and architecture

Step 2: Multiprocessing initialization (*__main__*)

- **Functionality:** Orchestrates the complex communication between the isolated components of the pipeline.
- **Code mapping:** Uses *multiprocessing.Value* and *multiprocessing.Event* to create synchronized states, such as *sys_state_val* (to share phase changes) and *sys_start_event* (allowing the operator to remotely trigger the participant's screen).

Pipeline mapping - Streaming and diagnostics

Step 3: Background hardware streamer (*start_stream*)

- **Functionality:** Bypasses Bluetooth scanning to lock onto the target device instantly.
- **Code mapping:** Runs as a daemonized process and evaluates *TARGET_MAC_ADDRESS* to execute the direct *stream()* function.

Pipeline mapping - Streaming and diagnostics

Step 4: Operator diagnostics view (*OperatorDiagnostics*)

- **Functionality:** Renders a real-time oscilloscope, assesses signal quality, and tracks the participant's event-related desynchronization (ERD).
- **Code mapping:** Uses environment variables (`os.environ['SDL_VIDEO_WINDOW_POS'] = "20,50"`) to forcefully push the Pygame window to the operator's monitor. It calculates cognitive density by **dividing live alpha and beta band power against the total power** (`self.alpha_pwr / total_pwr`) to render visual telemetry bars.

Tracking focus through ERD

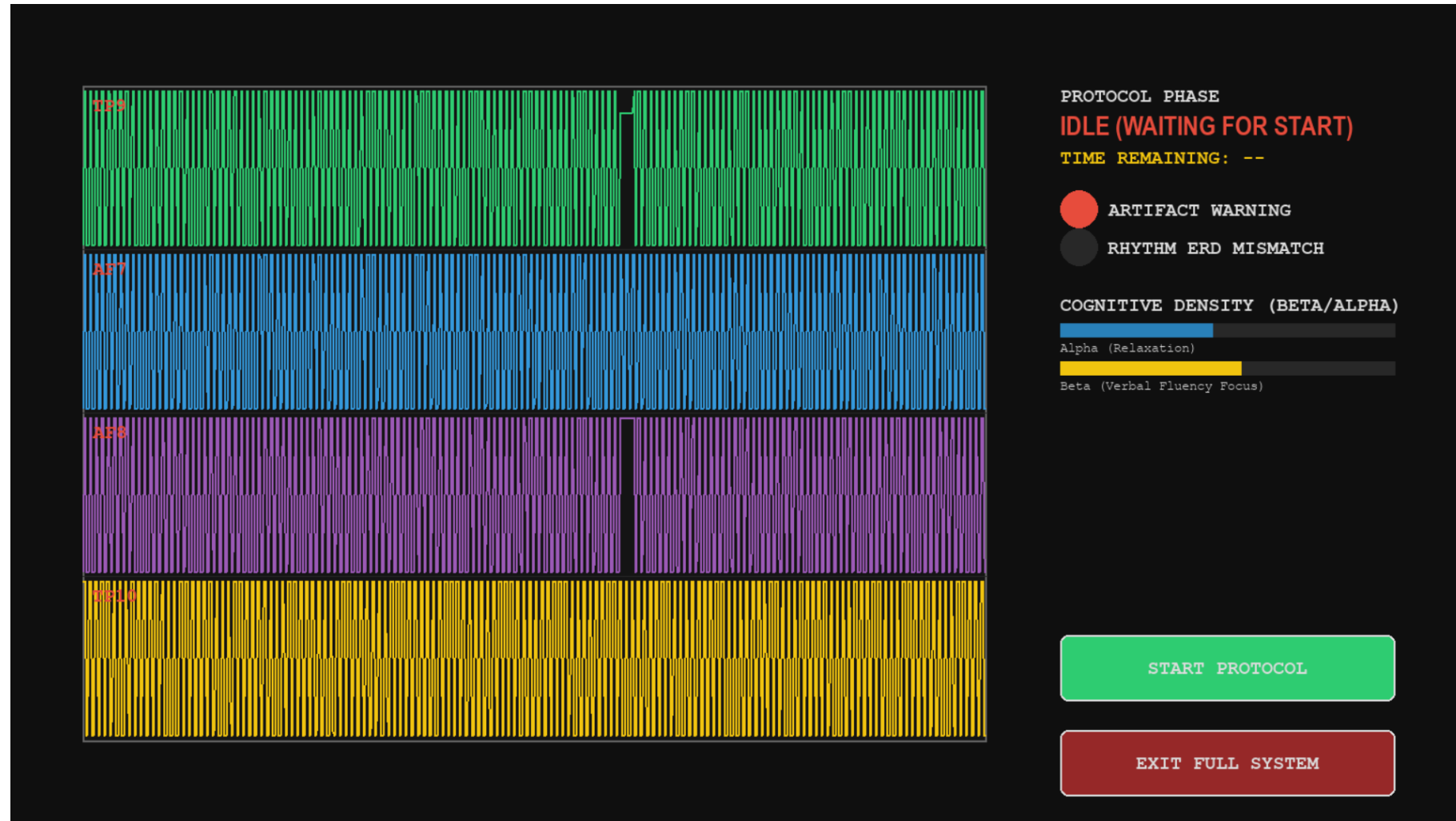
When your brain **shifts from relaxing to solving a problem**, its slow, idle brainwaves (**Alpha waves**) naturally drop in power.

During the resting phase, the system calculates your unique, personalized **Alpha baseline**.

When the **cognitive task** begins, the system continuously **compares your live Alpha levels against that baseline**.

If your Alpha power **fails to drop** (**remaining at or above 85% of your baseline**), the system assumes you have **lost focus** and triggers an "**ERD MISMATCH**" warning.

Operator's view



Pipeline mapping - Protocol execution of tasks

Phase 1 and 2 - Offline validation and the Berger effect

Phase 3 - Live calibration engine (The resting phase)

Phase 4 - Active cognitive load (Math vs. Fluency)

Unified diagnostic logic and live telemetry

Phase 1 and 2 - Offline validation and the Berger effect

Standardized baselines

- Sessions begin with 30-second "Eyes Open" and "Eyes Closed" phases.
- Auditory beeps guide the process to capture pure resting states without screen distraction.

The "Alpha block" phenomenon (The Berger effect)

- Eyes closed phase triggers a massive surge of idle Alpha waves.
- While eyes open instantly crashes Alpha levels, creating a massive physical signal shift that can overshadow subtle cognitive data.

Offline validation vs. machine learning

- **The offline script uses the "Eyes Closed" Alpha surge strictly to prove genuine sensor contact.**
- "Eyes closed" data is intentionally hidden from the machine learning model. The AI only compares "eyes open" rest against the active task, forcing it to learn actual cognitive effort rather than relying on the massive visual Alpha block.

BASELINE (EYES OPEN)

Stare at the crosshair and relax.

BASELINE (EYES CLOSED)

Close your eyes and relax completely.

Phase 3 - Live calibration engine (The resting phase)

Comparing apples to apples

- Requires an "Eyes Open (Relaxed)" baseline to prevent the physical "Alpha block" from corrupting cognitive data.

Dynamic neural calibration

- A 60-second resting phase (started by a 1200 Hz cue) that calculates a rolling average of your Alpha power.

Establishing the threshold

- Locks in a personalized baseline of your brain simply absorbing visual data without performing active tasks.

RESTING PHASE

Keep your eyes open and relax.

Phase 4 - Active cognitive load (Math vs. Fluency)

Task execution

- The final 60-second active phase is initiated by a 1500 Hz marker, at which point the protocols diverge to test different cognitive networks.

Complex mental math

- Deploys a rigid **$X \pm X \pm X$** equation format that refreshes every **4 seconds**. This rapid pacing creates a continuous cognitive strain to sustain Beta-band engagement.

ACTIVE FOCUS

Solve Mentally:
 $8 + 3 - 9 = ?$

The cloze task

- Participants view technical terms with a single missing character replaced by an underscore (e.g., **B A N D P _ S S**) and refresh the prompt every **2 seconds** to drive continuous working memory retrieval.

ACTIVE FOCUS

Mentally identify the missing letter:

B A N D P _ S S

Unified diagnostic logic and live telemetry

ERD enforcement

Both paradigms rely on the exact **same ERD logic**. The system continuously compares live Alpha power during the Phase 4 active task strictly against the Phase 3 "Eyes Open" Alpha power of the resting baseline.

Automated warning systems

If a participant's active Alpha power **fails to drop below 85%** of their established **baseline**, the operator's dashboard triggers a **visual mismatch** warning to flag a **lack of engagement**.

Threshold justification

The **85%** baseline threshold, requiring a 15% reduction in Alpha power, acts as a scientifically validated heuristic to filter out natural neural fluctuations and dry-sensor noise, ensuring the measured desynchronization reflects genuine cognitive effort.

Adaptive operator interfaces

The underlying telemetry architecture remains identical across both tasks, but the interface automatically adapts its cognitive density labels—displaying either "Beta (Mental Math Focus)" or "Beta (Verbal Fluency Focus)".

EXPERIMENTAL PROTOCOL: NEURAL VALIDATION, CALIBRATION, AND TASK EXECUTION

Phase 1 & 2 - Offline Validation (0-60s)



0-30 sec

+
BASELINE (EYES OPEN) |
Stare at the crosshair
and relax.



30-60 sec

—
BASELINE (EYES CLOSED) |
Close your eyes and relax
completely.

Data Visualization



Mixed-Amplitude Mixed-frequency
(Berger Effect)



1000 Hz
Cue



1000 Hz
Cue

Phase 3 - Live Calibration Engine (60-120s)



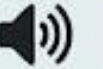
Avatar 3

RESTING PHASE |
Keep your eyes open
and relax.

Rolling Average Engine



System captures rolling live Alpha waves.
Locks-in to Apples to Apples Alpha Mean Threshold



1200 Hz
Cue

Phase 4 - Active Cognitive Task (120-180s)



Avatar A



Complex Mental Math

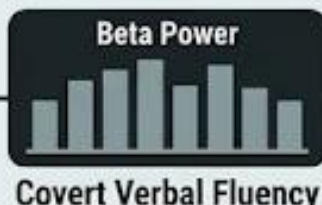
ACTIVE FOCUS |
Solve Mentally:
 $8 + 3 - 9 = ?$
Pacing: 4s Refresh ("Goldilocks" Zone)



1500 Hz
Cue



Avatar B



Covert Verbal Fluency

ACTIVE FOCUS |
Mentally identify the missing
letter:
F O _ U S
Pacing: 2s Refresh (Non-repeating letter generators)



1500 Hz
Cue

Unified Diagnostic Logic & Live Telemetry

Does Live Alpha (P4) <
85% Phase 3 Threshold?

Threshold Justification:
A 15% reduction filters out
natural neural fluctuations
and dry-sensor noise.



Cognitive
State



ERD / Rhythm
Mismatch

Added Justification:



Valid Cognitive State



ERD / Rhythm Mismatch

Cognitive Density Widgets



Beta Math Focus



Beta Verbal Focus

*Thank
You!*



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